RUCHITHA H K

• Bengaluru • ruchiikrish05@gmail.com • +91-9632451030 • www.linkedin.com/in/ruchitha-052066258

PROFESSIONAL SUMMARY

AI & Data Science graduate with a passion for innovation and a strong foundation in machine learning, deep learning, and image processing. Demonstrated success in leading impactful projects across healthcare, agriculture, and infrastructure sectors. Proven ability to leverage technical expertise to drive transformative solutions. Committed to continuous learning and contributing to the success of a forward-thinking organization focused on growth and innovation.

SKILLS

- **Programming Languages:** Python
- **Databases:** NoSQL, MongoDB
- **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Quantum Computing, Image Processing, Computer Vision, Explainable AI (XAI), Generative AI
- **Data Analytics:** Data Visualization (Tableau, PowerBI), Big Data, SQL

EXPERIENCE

Defect Detection in High Tension Insulators using Deep Learning
AI Intern | Compviz Technologies Private Limited, Bengaluru *April 2024 - May 2024*

- Leveraged drone footage to implement a deep learning-based method for automatic detection of flaws in high-tension insulators, resulting in improved efficiency and accuracy.
- Delivered valuable insights for power line infrastructure maintenance and inspection, boosting overall dependability and safety through advanced image processing and problem detection.

EDUCATION

Bachelor of Engineering (Artificial Intelligence and Data Science) *Dec 2020 - July 2024*
Global Academy of Technology, Bengaluru **CGPA 9.65/10 Percentage – 89%**

Class 12th – Karnataka State Board *March 2019 – March 2020*
Expert PU College, Mangalore **Percentage – 92.8%**

Class 10th - Karnataka State Board *April 2017 – April 2018*
Soundarya High school, Bengaluru **Percentage – 94.8%**

PROJECTS

Enhancing PCOS Detection and Classification *Nov 2023 - April 2024*

- Evolved an advanced diagnostic framework using diffusion models, GANs, zero-shot learning, and integrating transfer learning and explainable AI, enhancing accuracy and transparency in PCOS detection from ultrasound images.
- Focused on generating high-quality images with diffusion models and applying zero-shot learning techniques, contributing to robust and interpretable results that enhance early and precise PCOS detection.

Wheat Disease Classification and Prediction Using Deep Learning *June 2023 - Aug 2023*

- Utilised explainable AI tools and transfer learning techniques to enhance the interpretability and accuracy of the deep learning model developed for identifying fungal diseases in wheat plants.
- Led the evaluation of model performance, identified biases, and utilized explainable AI technologies to provide insights into prediction processes, enabling agricultural specialists to swiftly address and resolve wheat crop issues for improved health and yield.

Machine Learning Classifier for Breast Cancer Classification and Detection *Dec 2022 - Feb 2023*

- Designed an approach that provided valuable insights and meaningful conclusions, enhancing early and precise breast cancer diagnosis and improving survival rates through advanced ML methods.
- Oversaw the implementation process, ensuring precise analysis and evaluation of model performance, leading to valuable insights and improved breast cancer detection methodologies.

Machine Learning Approaches for Stroke Detection and Smote for Imbalanced Data *June 2022 - Aug 2022*

- Enhanced supervised machine learning algorithms to predict stroke conditions, incorporating the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance and improve accuracy for early detection.
- Presented a high-accuracy risk prediction model by analysing key stroke risk variables and implementing ML techniques, supporting timely prevention and detection of stroke in patients.

Evaluation of Unconfined Compressive Strength of Cement Using Ensemble Learning Regression *Dec 2021 - Feb 2022*

- Developed a robust predictive model using ensemble learning regression for assessing the unconfined compressive strength (UCS) of cement, enhancing cement design methodologies through data construction and exploratory data analysis (EDA)
- Applied advanced regression methods and validated results to improve the accuracy of UCS predictions, supporting more effective cement design practices.

INTERNSHIPS

Location-Constrained Virtual Network Embedding Project
Research Intern | National Institute of Technology, Karnataka *Sep 2023 - Dec 2023*

- Designed a novel solution for location-constrained virtual network embedding to minimize location constraints and reduce communication latency, utilizing an innovative greedy algorithm for optimal network resource utilization.
- Provided practical insights and recommendations for integrating location-aware approaches, contributing to improved network performance and efficiency in resource management.

CERTIFICATIONS

- Natural Language Processing Foundation Certification - Infosys Springboard Certification
- Cyber Security and Privacy - NPTEL Certification
- Privacy and Security in Online Social Media - NPTEL Certification
- Programming Foundations with JavaScript, HTML and CSS - Coursera Certification

ACHIEVEMENTS

Winners of All India Women Only Hackathon Organized by Shooting Star Foundation with the "Acadevo" Project - Acadevo is an educational website aimed at supporting 10th-grade students in rural areas. It offers access to past exam papers, comprehensive study resources, and information on available scholarships. The platform also features a personalised chatbot to provide tailored assistance and support for students' educational needs.

PUBLICATIONS

- Wheat Disease Classification and Prediction using Deep Learning - 2024 IEEE CONIT
- Machine Learning Classifier for Breast Cancer Classification and Detection - 2023 CIISCA
- Machine Learning Approaches for Stroke Detection - 2023 ERCICA
- Evaluation of Unconfined Compressive Strength of Cement Using Ensemble Learning Regression - 2022 IEEE CONECCT